Conditional RNA Sequence Design Using a Discrete Diffusion Model

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***Abstract*—RNA sequence design is an inverse-folding problem in which a nucleotide sequence must be generated to match a desired secondary structure. This project implements a discrete diffusion model for conditional RNA sequence generation and benchmarks it against ViennaRNA's classical inverse-folding algorithm, RNAinverse. The model uses an absorbing-state masking process, a Transformer-based denoiser, and classifier-free guidance. Experiments were conducted on 30 synthetic hairpin targets using two model configurations. RNAinverse achieved a 97% success rate with a mean defect of 0.087, while the diffusion models achieved success rates of 63% and 43%, respectively. Training loss analysis suggests that the model learned nucleotide composition but did not adequately capture complementary base-pairing relationships. Sampling and evaluation limitations are also identified, motivating improvements in structural conditioning, validation, and stochastic generation.**

***Index Terms*—RNA sequence design, Discrete diffusion, Transformer, RNA secondary structure, Classifier-free guidance**

I. Introduction

RNA sequence design is an inverse-folding problem: given a desired secondary structure, the objective is to identify a nucleotide sequence that folds into it. Unlike conventional structure prediction, which maps sequences to structures, inverse folding operates from the target structure to a desired sequence. Diffusion models provide a generative framework for this task by progressively reconstructing data from corrupted representations. This project explores a discrete diffusion approach for conditional RNA sequence generation, using secondary structure as conditioning information. The model is benchmarked against ViennaRNA's classical inverse-folding algorithm, RNAinverse. The primary evaluation criterion is whether the generated sequence folds into the specified target structure.

II. Methods

A. Evaluation and Benchmark

An evaluation harness (`eval_harness.py`) was implemented using ViennaRNA. It provides four functions: MFE folding, exact target-structure matching, normalized ensemble defect calculation, and aggregate evaluation of success rate and mean ensemble defect. Basic tests were performed using correctly and incorrectly folding sequences.

A benchmark of 30 synthetic hairpin structures was generated using random seed 0. Stem lengths ranged from 3–8 base pairs and loop lengths from 3–6 unpaired nucleotides. RNAinverse was used to design sequences from random initial sequences, establishing the classical baseline.

B. Discrete Diffusion Model

The forward process uses absorbing-state masking: each nucleotide is independently replaced by a MASK token with probability t, sampled uniformly from $[0, 1]$. A Transformer encoder predicts the original nucleotide identities at masked positions.

The model uses 4 or 6 Transformer layers, dimension 128, and 4 attention heads. Input embeddings combine nucleotide tokens, secondary structure tokens, positional information, diffusion time, and, in the second configuration, base-pair partner indices. The output consists of logits over A, C, G, and U.

Training sequences of lengths 12–24 were randomly generated and folded using ViennaRNA. Only sequences containing at least one base pair were retained. Cross-entropy loss was calculated exclusively at masked positions. Classifier-free guidance was implemented by replacing structural conditioning with a null token for 10% of training examples. During sampling, conditional and unconditional logits were combined as

$$l_g = l_u + w(l_c - l_u), \quad w = 2.0 \tag{1}$$

Generation begins with an all-MASK sequence and proceeds over 12 steps, revealing the most confident masked positions at each step. The number of positions revealed per step is $k = \lceil \frac{L}{12} \rceil$, where L is the sequence length.

C. Experimental Configurations

Three experiments were conducted:

- **Run A:** RNAinverse on the 30-target benchmark.
- **Run B:** Four-layer Transformer, no partner features, 20,000 training pairs, 10,000 training steps, and guidance strength 2.0.
- **Run C:** Six-layer Transformer with partner-index embeddings, 50,000 training pairs, 30,000 training steps, cosine learning-rate decay, weight decay 0.01, gradient clipping at 1.0, and guidance strength 2.0.

III. Results

A. Benchmark Performance

Table I summarizes the performance of the three configurations.

TABLE I
PERFORMANCE ON THE 30-TARGET HAIRPIN BENCHMARK.

Method	Configuration	Success	Mean defect
RNAinverse	Classical baseline	97% (29/30)	0.087
Diffusion B	4 layers, no partners	63% (19/30)	0.168
Diffusion C	6 layers, partners	43% (13/30)	0.287

RNAinverse substantially outperformed both diffusion configurations. Run B achieved moderate performance, whereas Run C showed a regression despite increased model depth, training data, and training duration. RNAinverse required approximately 0.001 seconds per target. An illustrative failure involved target `(((.....)))`, for which the generated sequence `AUCGUCUCCGGU` did not reproduce the target MFE structure and had an ensemble defect of 0.325.

B. Training Loss

Run C's training loss decreased from 1.5119 initially to 1.2360 after 1,000 steps, but subsequently fluctuated between approximately 1.16 and 1.25 until the final recorded step (29,000: 1.1736). For reference, random guessing among four nucleotide bases gives a cross-entropy loss of $\ln(4) \approx 1.386$. Each reported value was calculated from a single batch of 128 sequences, with approximately +/- 0.03 variation attributable to batch noise.

The plateau suggests that the model learned basic nucleotide composition but did not adequately capture pairing relationships. This interpretation remains a hypothesis requiring further experiments.

IV. Discussion and Limitations

Run C underperformed Run B, but multiple variables were changed simultaneously, preventing attribution of the regression to any individual modification. One possible explanation concerns the partner-index representation: it identifies the position paired with each nucleotide but does not directly provide the nucleotide identity at that position. The Transformer must learn this relationship indirectly through attention.

The current implementation also has several limitations:

- **Deterministic sampling:** Greedy nucleotide selection produces identical samples, making the reported best-of-10 result invalid. Stochastic sampling has been identified as a necessary correction but is not yet implemented.
- **No validation split:** All training data was used for optimization, preventing independent assessment of generalization.
- **Single random seed:** Run-to-run variability remains unknown.
- **Training oracle dependence:** ViennaRNA generated the training structures and was also used for evaluation.
- **Oracle asymmetry:** RNAinverse directly optimizes against the same energy model used for evaluation, favouring the classical baseline.
- **Small benchmark:** With only 30 targets, each successful design changes the success rate by approximately 3.3 percentage points.

V. Future Work

The next step is to implement stochastic sampling for best-of-N evaluation while retaining greedy sampling for single-sample comparisons. A validation split containing 5,000 held-out sequence-structure pairs will then be introduced, with validation loss recorded every 1,000 training steps. Controlled ablation experiments will isolate the effect of partner-index features. A partner-sequence embedding, explicitly providing each position with its partner's current nucleotide, will also be investigated to improve the representation of base-pairing relationships. Further experiments will include stem-length stratification, guidance-strength evaluation ($w = 0, 1, 2, 4$), and multiple random seeds. Finally, the model will be extended to longer structures and external benchmarks such as Eterna100 and RNA-Puzzles, with success rate compared against RNAinverse across sequence lengths.

VI. Conclusion

This project establishes a preliminary framework for conditional RNA sequence generation using discrete diffusion. Although the current models underperform RNAinverse, the experiments highlight the difficulty of learning structural dependencies from masked nucleotide sequences. The training-loss plateau and performance regression motivate improvements in partner information representation and sampling. Future work will focus on controlled ablations, validation, stochastic generation, and evaluation on larger external benchmarks.